EXAMINATION: Computed Tomography Angiography (CTA) of the Chest for Pulmonary Embolism.

INDICATION: A 59-year-old female, postoperative day 4 following left total hip arthroplasty, presenting with sudden-onset tachycardia, pleuritic chest pain, and SpO2 desaturation to 88%.

TECHNIQUE: Volumetric axial helical CT images of the thorax were obtained from the lung apices to the adrenal glands following the intravenous administration of 75 mL of Isovue-370 contrast via an 18-gauge right antecubital catheter using a dual-barrel power injector. Bolus tracking was triggered in the main pulmonary artery at 120 Hounsfield units. Multiplanar sagittal, coronal, and maximum intensity projection (MIP) reconstructions were generated.

FINDINGS:

Pulmonary Vasculature:

- The main pulmonary artery caliber is normal, measuring 26 mm.

- Extensive, occlusive thromboembolic filling defects are visualized within the right interlobar pulmonary artery, extending into the right middle lobe and right lower lobe superior and basilar segmental branches.

- Additional non-occlusive filling defects are noted within the left lower lobe posterior basal segmental pulmonary artery.

- The left upper lobe pulmonary arterial branches are patent without emboli.

Right Ventricular Strain Assessment:

- The right ventricle-to-left ventricle (RV/LV) diameter ratio measured on axial reformatted four-chamber views is 1.15 (abnormal > 1.0, indicating mild acute right heart strain).

- The interventricular septum remains flat without paradoxical bowing into the left ventricle.

- Mild reflux of dense contrast medium is visualized into the inferior vena cava and suprahepatic veins.

Thoracic Cavity and Lungs:

- There is a wedge-shaped, pleura-based consolidation measuring 2.4 x 1.8 cm in the posterior basilar segment of the right lower lobe with surrounding ground-glass halo, characteristic of a pulmonary infarction (Hampton's hump).

- Small reactive right pleural effusion noted. No pneumothorax.

- Mediastinal and hilar lymph nodes are not pathologically enlarged. Normal cardiac silhouette.

- Visualized upper abdomen shows normal liver and adrenal glands.

IMPRESSION:

1. Acute bilateral pulmonary thromboembolism with high clot burden involving the right main/interlobar and segmental branches and left lower lobe segmental branches.

2. Evidence of acute right ventricular strain (RV/LV ratio 1.15 with hepatic contrast reflux).

3. Peripheral right lower lobe pulmonary infarction.